

Priyadarshan

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RESEARCH INTEREST

Focused on characterizing and mitigating the impacts of electrification on distribution grids using distributed energy resources (DER).

EDUCATION

Purdue University <i>PhD in Mechanical Engineering</i> Advisor: Dr. Kevin J. Kircher CGPA: 3.84/4.00	West Lafayette, IN, USA Jan 2022 – Present
Indian Institute of Technology (ISM) <i>MTech in Mechanical Engineering</i> Advisor: Dr. Subrata K. Ghosh CGPA: 9.50/10.00	India Aug 2019 – May 2021
Manipal Institute of Technology <i>BTech in Mechanical Engineering</i> CGPA: 9.38/10.00	India Aug 2013 – May 2017

RESEARCH EXPERIENCE

Graduate Research Assistant <i>Purdue University</i> Thesis Projects: • EDGE: Emulating the Distribution Grid Impacts of Electrification – Developed a simulation test-bed for investigating residential and vehicle electrification impacts on distribution grids. Estimates transformer loss of life, power quality, and DER optimization. DERs reduce peak demand by 35% and transformer degradation by 99%. Published in HICSS, 2024 • Distribution Grids as a Barrier to Residential Electrification – Quantified how distribution infrastructure limits U.S. residential electrification by modeling three million homes and six million electric vehicles across 3000+ counties. Coldest regions see 5× today's peak demand; full electrification needs 600 GW at \$350–\$790B nationally; demand-side management cuts two-thirds of costs. Published in Cell Reports Sustainability, 2025. • First Nationwide Vehicle-To-Home (V2H) Study on Distribution Grid Impacts – First nationwide study on V2H discharging impacts on U.S. distribution grids. V2H halves grid reinforcement costs of residential and commercial electrification, offering a cost-effective pathway to grid resilience. Under review • Maximizing Untapped Grid Capacity for Affordable Electricity – Investigating how DERs unlock latent grid capacity to raise load factor, grid utilization and reduce electricity costs for residential consumers without grid build-out. <i>Manuscript in preparation</i>.	June 2022 – Present West Lafayette, IN, USA
Research Intern <i>National Laboratory of the Rockies</i> • Developed a co-simulation framework for commercial rooftop packaged units.	May 2025 – July 2025 Golden, CO, USA

OTHER COLLABORATIVE PUBLICATIONS

- E.N. Pergantis, L.D. Premer, A.H. Lee, H. Liu, **Priyadarshan**, et al. Protecting residential electrical panels and service through model predictive control: A field study, [Applied Energy, 2025](#).
- E.N. Pergantis, L.D. Reyes Premer, A.J. Khabbazi, **Priyadarshan**, et al. Active current limiting control of residential appliances for breaker panel protection across the US: a preliminary study, [CISBAT, 2025](#).
- E.N. Pergantis, **Priyadarshan** et al. Field demonstration of predictive heating control for an all-electric house in a cold climate, [Applied Energy, 2024](#).

AWARDS & FELLOWSHIPS

- Runner-up, Computational Interdisciplinary Graduate Programs Spring Symposium Doctoral Talk, Purdue (2026)
- Bilsland Dissertation Fellow (2026) | Perry Graduate Fellowship (2025) | William Uffman Fellowship (2024), Purdue University
- 1st Position – Jump Into STEM Challenge, National Building Science Competition, NLR (previously NREL) (2024)
- People's Choice Say It in 6 Award, Purdue (2023) | Gold Medal – Ranked 1st in MTech, IIT Dhanbad (2021)
- Post-Graduate Research Fellowship, IIT Dhanbad (2019) | 99th Percentile – GATE (2019) | Top 5% in BTech, Manipal (2017)

INVITED TALKS & POSTER PRESENTATIONS

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| • ME597: Distributed Energy Resources, Mechanical Engineering, Purdue University, USA | 2026 |
| • Impacts of Residential Electrification, Indiana Sustainability and Resilience Conference, IU Environmental Resilience Institute, USA | 2026 |
| • Impacts of Home and Vehicle Electrification on Distribution Grids, Grid of Tomorrow Consortium, Purdue University, USA | 2025 |
| • Griffin Lab, University of Colorado Boulder, USA | 2024 |

IN THE NEWS

- The cost (and savings) of electrifying homes and cars, AP segment (Lafayette, Evansville, Monterey-Salinas, San Francisco), [News](#) | Purdue offers solutions to blackouts as people electrify homes and cars, [IPM News](#) 2025
- If every home and vehicle went electric, power outages could spike, [Purdue News](#) | What electrification means for home and car owners, [Star City TV](#) | Electric homes could cost households thousands, [WISH-TV](#) 2025
- Let go of the thermostat: predictive heating control system demonstrates huge energy savings, [Purdue News](#) 2024

SKILLS, OUTREACH & SERVICE

- Programming & Tools:** C++, MATLAB, Python, Github, Tableau, Simulink, **Research:** Residential electrification, distribution grids, heat pumps, EVs, DERs, load growth, grid resilience, optimal control, building-grid interaction modeling | **Communication:** Undergraduate and graduate instruction, peer mentoring, technical writing, invited talks, media engagement
- Co-chair, Buildings Modeling subtrack, Herrick Conference (2026) | Vice-President, Herrick Student Resource Committee, Purdue (2024) | Social Media, Sports & Well-being Chair, OMEGA, Purdue (2023) | Volunteer, Minority Engineering Program | Placement & Internship Representative, IIT Dhanbad (2019–2021)